

PAVAN SAI GUNDARAM

Denton USA | +1 (940) 843-5105 | pavansaigundaram679@gmail.com | <http://www.linkedin.com/in/pavan-sai-gundaram>

SUMMARY

Experienced Data Scientist with 3+ years of experience leveraging statistical analysis and advanced ML/AI techniques to drive data-informed business decisions. Skilled in Python, SQL, and cloud platforms (AWS, Azure, GCP), with hands-on expertise in MLOps, transformers, LLMs, and deep learning. Proven ability to develop scalable AI solutions and optimize machine learning workflows for production environments.

EDUCATION AND TRAINING

MASTER OF SCIENCE: DATA SCIENCE

May 2024

University of North Texas , GPA: 3.75

BACHELOR OF TECHNOLOGY: COMPUTER SCIENCE

June 2022

TKR College of Engineering and Technology, GPA: 8.71

TECHNICAL SKILLS

- **Methodologies and IDEs:** SDLC, Agile, Visual Studio Code, PyCharm, Colab.
- **Languages & Databases:** Python, R, SQL, Scala, MATLAB, MongoDB, Redshift, MySQL, Oracle, Pyspark, Snowflake, Databricks.
- **ML & AI Frameworks:** TensorFlow, PyTorch, Keras, XGBoost, LightGBM, Scikit-learn, Hugging Face, LangChain, MLflow.
- **Cloud Platforms:** AWS (SageMaker, Lambda, Redshift, CloudWatch), Azure (Azure ML, Data Factory, Synapse), GCP (BigQuery).
- **Tools & Libraries:** Docker, Kubernetes, Spark, Hadoop, Airflow, Tableau, Power BI, Jenkins, CI/CD, NLTK, Terraform, MLOps.
- **ML Algorithms:** Regression, Supervised Learning, Unsupervised Learning, Random Forest, Linear Regression, Decision Tree, Deep Learning, Clustering, Classification, Time-Series, Open AI, LLM.
- **Advanced Techniques:** NLP, Transformers, GPT, Retrieval-Augmented Generation (RAG), GANs, CNNs, RNNs, model explainability (SHAP, LIME), A/B testing, feature engineering, hyperparameter tuning, Recursive Feature Elimination (RFE), Principal Component Analysis (PCA).
- **AI Search & Retrieval:** Dense Passage Retrieval (DPR), FAISS, Pinecone, Elasticsearch.
- **Data Engineering & Processing:** ETL/ELT Pipelines, Apache Spark, Hadoop, Airflow, Azure Data Factory, AWS Glue, SQL-based ETL, Feature Engineering, Data Transformation, Data Warehousing.

EXPERIENCE

KOFAX

USA

Data Scientist

May 2023 to Current

- Created and deployed a hybrid recommendation system utilizing collaborative filtering, content-based filtering, autoencoders, and LLMs, increasing sales by 15% and customer engagement by 12% by aligning product suggestions with user preferences.
- Increased recommendation precision by 20% through K-means clustering for customer segmentation, enabling targeted marketing strategies based on behavioral and preference data.
- Engineered a Retrieval-Augmented Generation (RAG) system with semantic search and fine-tuned LLMs, reducing AI resolution time for customer queries by 30% while achieving 95% accuracy.
- Achieved 92% prediction accuracy by optimizing ML models through feature engineering, cross-validation, and hyperparameter tuning using Python, Pandas, NumPy, and Scikit-learn.
- Automated ML model deployment with MLOps pipelines, integrating A/B testing, CI/CD (GitHub Actions), and automated retraining, reducing model drift by 25% and increasing user engagement by 10%.
- Designed and deployed scalable ML pipelines on AWS, leveraging SageMaker, Lambda, and Redshift. Automated ETL/ELT with Snowflake, Databricks, and Spark for streamlined model deployment.
- Implemented real-time model monitoring, using CloudWatch, Prometheus, and Airflow to minimize downtime. Optimized AI search with FAISS, Pinecone, and Elasticsearch, boosting retrieval speed by 40% and accuracy by 15%.
- Accelerated AI deployment by 40%, automating infrastructure with Terraform, GitHub Actions, and Kubernetes. Integrated NLP and LLMs for sentiment analysis, enhancing AI recommendations..

SIGMA INFO TECH

India

Machine Learning Engineer

January 2021 to June 2022

- Developed predictive models for property sale prices using machine learning algorithms and deep learning frameworks, improving prediction accuracy by 15% through advanced hyperparameter tuning.
- Designed and implemented ETL processes in ADF and Databricks, ensuring seamless data extraction, transformation, and loading, increasing data accessibility for modeling by 30%.
- Conducted extensive data preprocessing on real estate text data, reducing missing values by 5% through imputation and normalization while enhancing features using NLP techniques.
- Engineered and deployed ML models on Azure with high availability and disaster recovery solutions, improving system resilience and reducing downtime by 25%.
- Automated ML workflows and pipelines using Azure ML, Airflow, Terraform, and GitHub Actions, reducing deployment time by 40% and streamlining large-scale operations with CI/CD.
- Optimized model performance with Recursive Feature Elimination (RFE) and Principal Component Analysis (PCA), enhancing computational efficiency while maintaining predictive accuracy.
- Developed interactive dashboards with Tableau, Seaborn, Matplotlib, and Plotly, and deployed real-time property pricing models via Flask and Docker for seamless integration with web and mobile platforms.
- Improved operational efficiency and scalability by integrating Snowflake, Apache Spark, and Databricks, enabling real-time analytics, A/B testing, and ML model execution at scale.

CERTIFICATIONS

AWS Certified: Data Engineer Associate